Future Tech:-

Sign Language to Text or Speech Translator



Name:-Anwita Datta

Idea Title

The Relevance of Indian Sign Language in Today's World.

1.Data Preprocessing:-

- Image/Video Processing: Capture high-quality video footage of the sign language user.
- Segmentation: Isolate the hands and facial expressions from the background to focus on the features for language recognition.

2.Sign Recognition:-

- Machine Learning Models
- Gesture Recognition Algorithms

3.Contextual Analysis

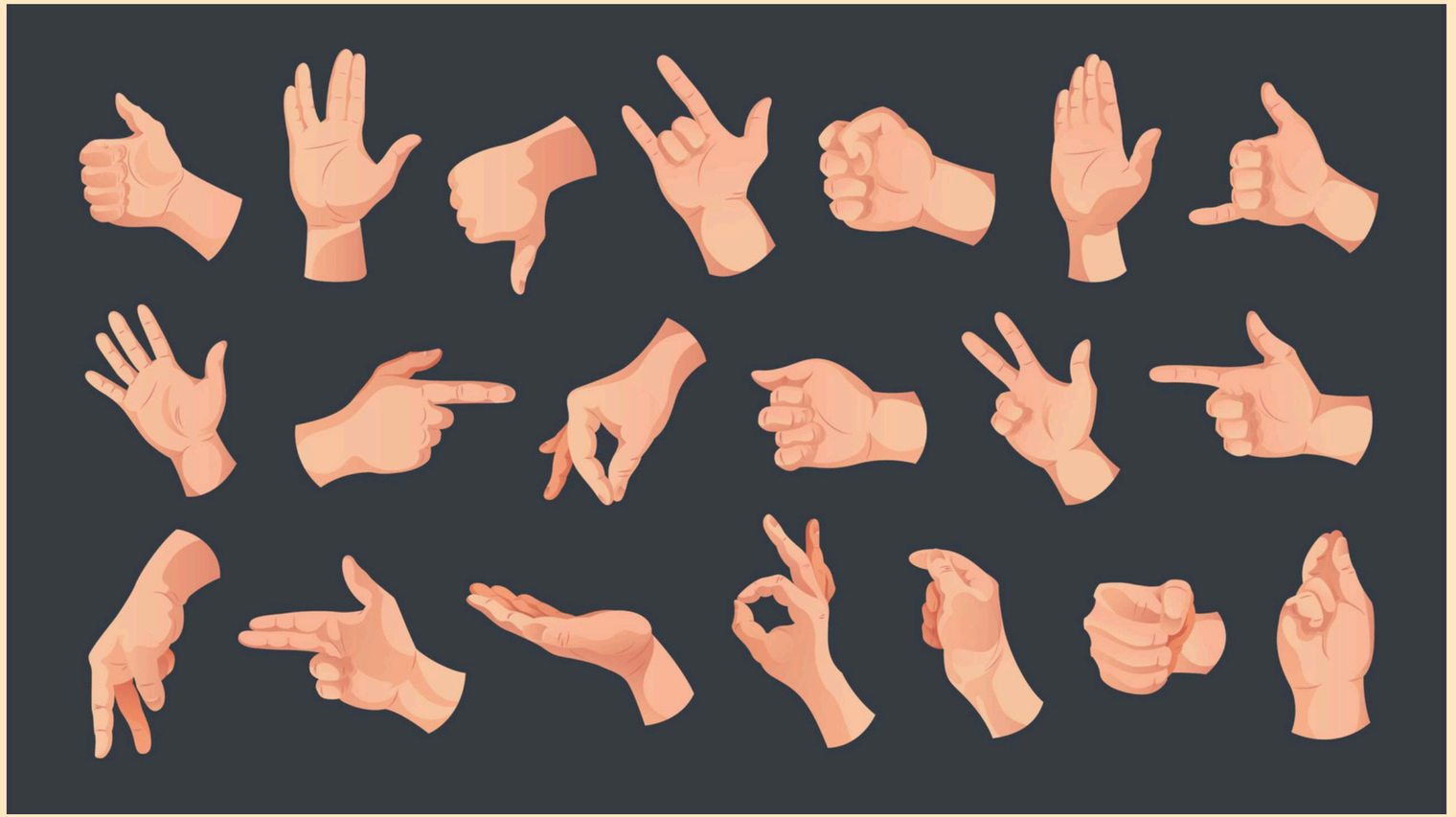
- Language Modelling
- Syntax and Grammar Correction

4.Real-Time Translation

- Text Generation
- Error Correction

5.User Interface and Display

- Real-Time Text Display
- Interactive Feedback



Technical Approach

Data Requirements:-

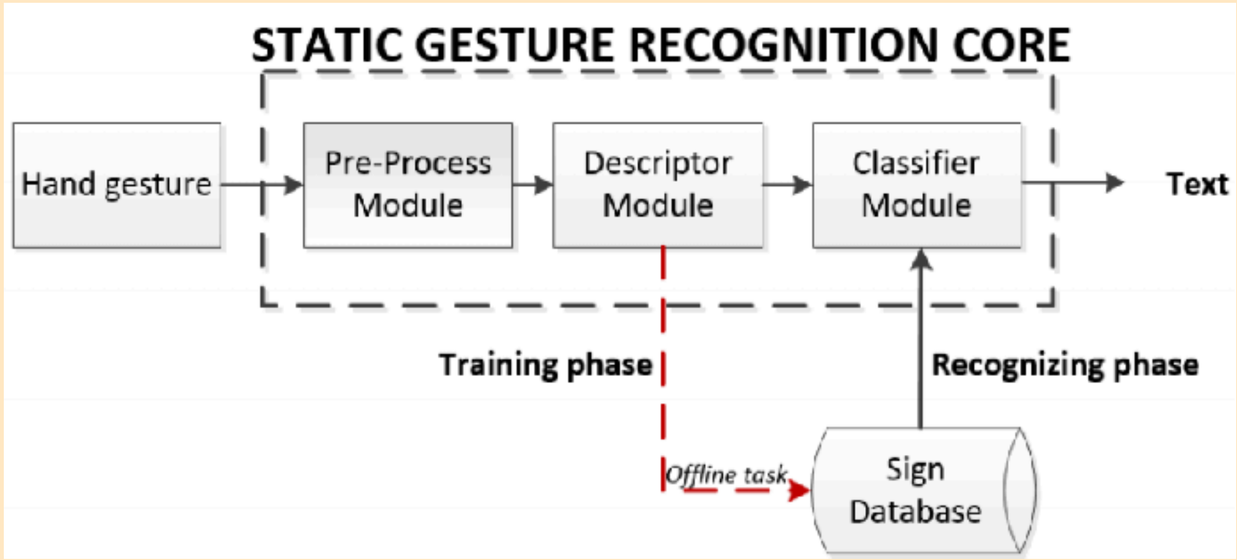
- Training Datasets: Labelled ISL Datasets/Data Augmentation
- Networking and Cloud Infrastructure
- Real-Time Communication Protocols

Hardware Requirements:-

- High Resolution Cameras with high Frame Rate-+
- Processing Unit:GPU/RAM/CPU
- Sensors and Wearables

Software Requirements :-

- Operating System: Linux or Windows
- Computer Vision Libraries: OpenCV/Media pipe
- Deep Learning Frameworks: TensorFlow/PyTorch:keras
- Natural Language Processing(NLP) Tools: NLTK/Transformers/Spacy
- Real-Time Data Processing Tools: Apache Kafka/Flume.



Feasibility and Viability

1.Exclusion of Non-Standard Signs:

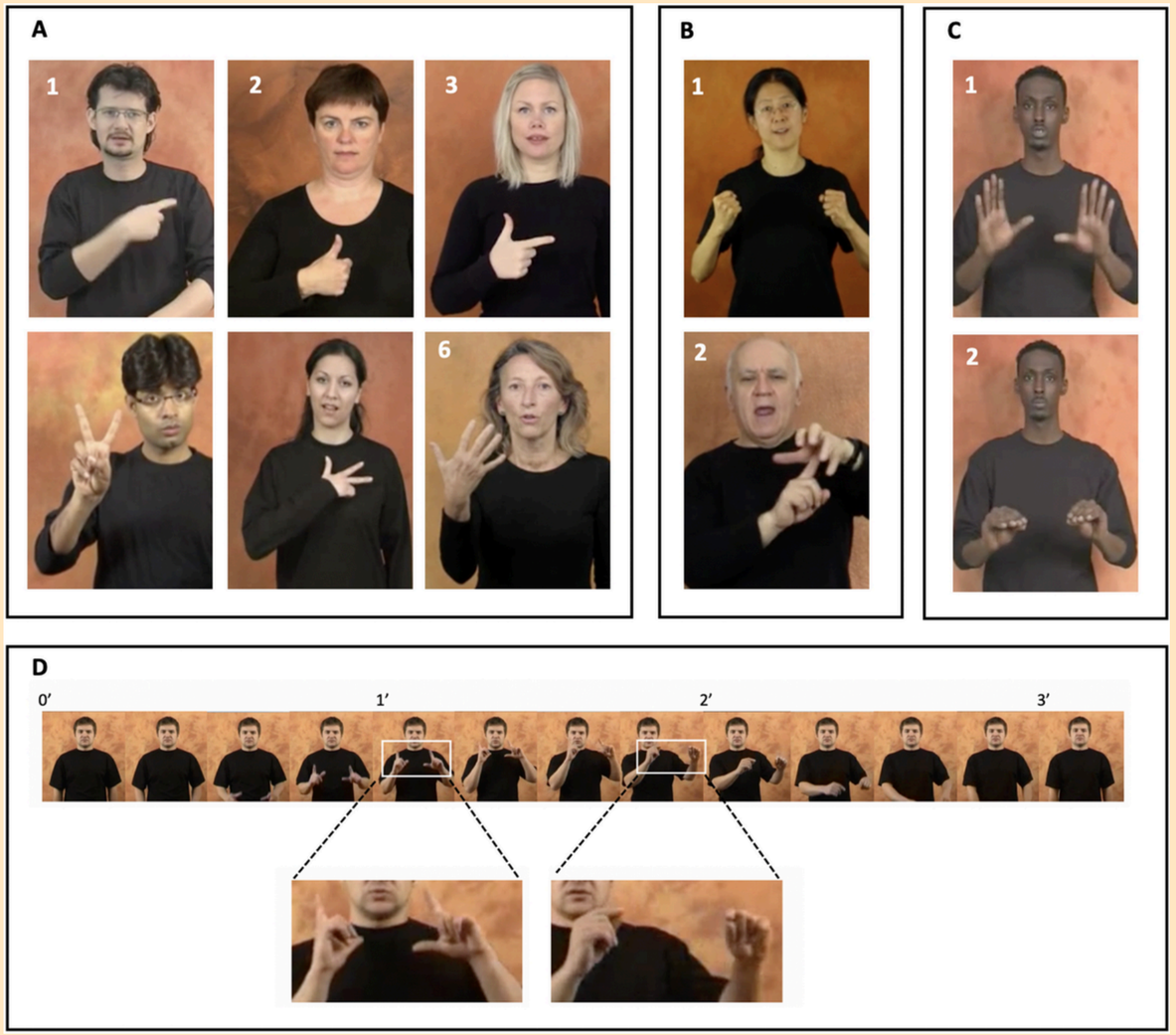
- Dialectical Variations
- Customized Signs
- (can be overcome by manual data updation)

2.Accuracy Issues:-

- Misinterpretation
- Incomplete Translations
- (can be overcome by contextual understanding

3.Loss of Nuance and Meaning:

- Context and Emotion
- Cultural Significance
- (can be overcome by emotional annotation)



Impact and Benefits

Impacts:-

- Enhanced Communication
- Workplace Integration
- Inclusion in Education
- Accessibility
- Empowerment of Deaf Advocacy Groups

Benefits:-

- Technological Innovation
- Educational Benefits
- Increased Independence
- Improved Public Services
- Strengthening Community Networks

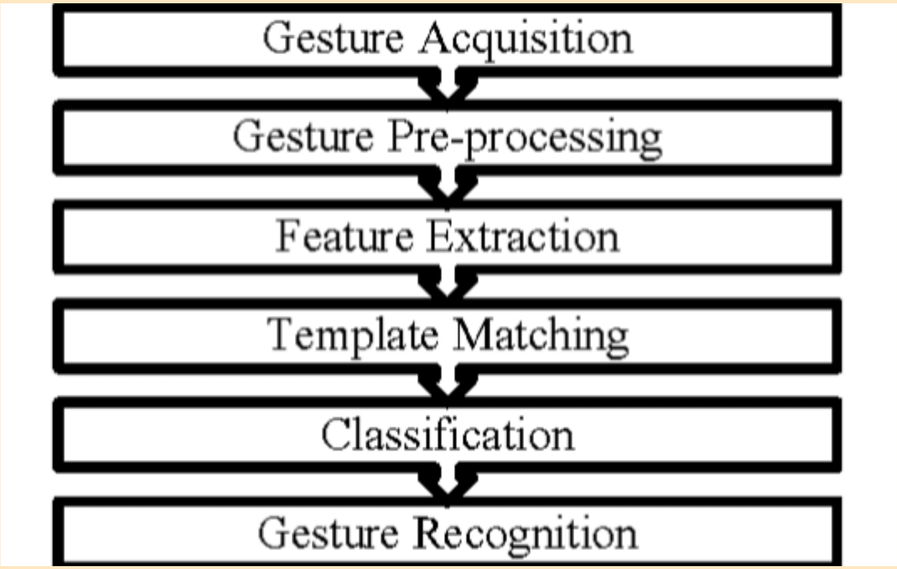


Research and References

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Thank You!

